
RESEARCH INTERNSHIP REPORT

Benchmarking ASR and Speech Translation Models for Low-Resource Languages Spoken by Asylum Seekers



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Contents

1 Overview and Objectives	2
1.1 Languages Covered	2
2 Datasets Used	3
2.1 Dataset Selection Considerations	3
2.2 Task-Specific Dataset Considerations	4
3 Evaluation Metrics	4
3.1 Automatic Speech Recognition Metrics	4
3.2 Speech Translation Metrics	5
4 Medium-Low Resource Language-Specific Benchmarks	5
4.1 Arabic	5
4.2 Persian	5
4.3 Georgian	6
4.4 Russian	6
4.5 Somali	6
4.6 Tigrinya	7
4.7 Turkish	7
5 Model-Wise Systematic Benchmarks	7
5.1 TranslateGemma Benchmark	7
5.2 TranslateGemma Findings	8
6 Whisper Architectural Benchmark	9
6.1 Whisper Observations	10
7 Gemma 4 12B Prompt Evaluation	10
7.1 Prompt Configuration	10
8 Omnilingual ASR Benchmark	12
9 Key Findings and Limitations	17
9.1 Dataset and Training-Data Overlap	17
9.2 Long-Form Speech Limitations	17
9.3 COMET-BLEU Metric Disagreement	17
10 Comparative Model Observations	18
10.1 Whisper	18
10.2 Gemma 4	18
10.3 OmniASR	18
11 Future Work	18
12 Conclusion	19

1 Overview and Objectives

This report presents a systematic evaluation of Automatic Speech Recognition (ASR) and Text-to-Text Translation models for low-resource languages spoken by asylum seekers. The study is motivated by the potential use of these technologies in asylum and legal proceedings, where transcription or translation errors may affect the interpretation of an individual's testimony. The primary objectives of the internship were to:

1. **Benchmark ASR and speech translation models** across multiple low-resource languages.
2. **Identify model errors and failure modes**, including omissions, mistranslations, hallucinations, and inconsistent outputs.
3. **Assess the reliability of these models** for potential use in high-stakes asylum and legal settings.

1.1 Languages Covered

The benchmark covers **21 languages** spanning diverse linguistic and geographic contexts.



2 Datasets Used

The benchmark was constructed using publicly available speech and text datasets from multiple sources, including Hugging Face, Common Voice, VoxForge, and other publicly released corpora. The dataset and split used for each language are summarized below.

#	Language	Dataset / Corpus	Split
1	Somali	DDD-Kenya/Somali-ASR-Subset-68H	Train
2	Italian	VoxForge – Italian	–
3	Russian	VoxForge – Russian	–
4	Tamil	deepdml/iisc-mile-tamil-asr	Test
5	Luxembourgish	disco-eth/WorldSpeech	Train
6	Central Kurdish	Common Voice Scripted Speech 26.0 – Central Kurdish	ASR-Only
7	Northern Kurdish	Common Voice Scripted Speech 26.0 – Kurmanji Kurdish	–
8	Persian	pourmand1376/asr-farsi-youtube-chunked-30-seconds	Train
9	Arabic	disco-eth/WorldSpeech	Test
10	Georgian	shunyalabs/georgian-speech-dataset	Train
11	Tigrinya	google/WaxaINLP	Train
12	Turkish	VoxForge – Turkish	–
13	Afghan Dari	RFE/RL Afghan Dari News Text Corpus	Text-to-Text
14	Albanian	Common Voice Scripted Speech 26.0 – Albanian	–
15	Irish	Common Voice Scripted Speech 26.0 – Irish	–
16	Romansh	Common Voice Scripted Speech 26.0 – Romansh Sursilvan	–
17	Romanian	Common Voice Scripted Speech 26.0 – Romanian	–
18	Malayalam	Common Voice Scripted Speech 26.0 – Malayalam	–
19	English	Common Voice Spontaneous Speech 4.0 – English	–
20	French	Common Voice Spontaneous Speech 4.0 – French	–
21	Spanish	Common Voice Spontaneous Speech 4.0 – Spanish	–

Table 1 : Datasets used for the multilingual benchmark

2.1 Dataset Selection Considerations

The objective was to obtain a consistent and manageable evaluation set across 21 languages. Accordingly, the following sampling strategy was adopted:

1. **Hugging Face datasets:** For datasets hosted on Hugging Face, the available dataset split was first identified according to the dataset specification. Depending on the dataset, either the train or test split was used. After fixing the split, the **first 30 audio samples** were collected for evaluation. This procedure was applied consistently across the selected Hugging Face datasets.

2. **VoxForge and Common Voice datasets:** For VoxForge and Common Voice datasets, where a specific train/test split was not used for the selected data, the corresponding language-specific dataset was first downloaded. Following download, **30 audio samples were randomly selected** from the available data. Random sampling was adopted to avoid dependence on the ordering of the downloaded files and to obtain a representative subset.

2.2 Task-Specific Dataset Considerations

The availability of appropriate speech and translation resources, as well as the language coverage of the evaluated models, was taken into account when constructing the benchmark. Consequently, all languages could not be evaluated across both the automatic speech recognition (ASR) and speech translation (ST) tasks. The following task-specific considerations were applied:

1. **Afghan Dari:** The Afghan Dari dataset contains only textual data and does not provide speech recordings. Consequently, it is used exclusively for the **speech translation evaluation** as a text-to-text translation resource and is not included in the ASR evaluation.
2. **Central Kurdish:** Central Kurdish is not supported by most of the speech translation models evaluated in this study. Therefore, it is included exclusively in the **ASR evaluation** and omitted from the speech translation comparison.
3. **English:** For the speech translation evaluation, the source-language text for each language is translated into **English**, which serves as the common target language across the benchmark. Since English is already the target language, **English is excluded as a source language** from the speech translation evaluation.

3 Evaluation Metrics

3.1 Automatic Speech Recognition Metrics

The primary ASR metrics used were:

- **Word Error Rate (WER):** Measures word-level substitutions, deletions, and insertions with respect to the total number of words in a sentence.
- **Character Error Rate (CER):** Measures errors at the character level. It is particularly useful for languages where word segmentation may not be reliable.
- **Match Error Rate (MER):** Measures word-level recognition errors while considering substitutions, deletions, insertions, and correctly matched words instead of total number of words.
- **Word Information Lost (WIL):** Denotes the amount of word-level information lost between the reference and transcribed texts.

Lower values indicate better ASR performance for WER, CER, MER, and WIL.

3.2 Speech Translation Metrics

Translation quality was evaluated using:

- **BLEU**: Evaluates overlap between translated and reference texts using n -gram precision.
- **COMET**: A learned evaluation metric that considers source text, reference translation (Google Translated texts of source texts), and model translated texts, allowing it to capture semantic similarity beyond exact n -gram overlap.

For BLEU, unigram, bigram, trigram, and four-gram overlaps were also examined individually:

- 1-gram: Overlap of individual words.
- 2-gram: Overlap of consecutive two-word sequences.
- 3-gram: Overlap of consecutive three-word sequences.
- 4-gram: Overlap of consecutive four-word sequences.

For TranslateGemma, the COMET checkpoint used was **Unbabel/wmt22-comet-da**. COMET values are dependent on the specific checkpoint used; therefore, results obtained using different COMET checkpoints may differ.

4 Medium-Low Resource Language-Specific Benchmarks

Language-specific exploratory benchmarks were developed to closely analyse the behaviour of speech models on selected low-resource and medium-resource languages. **Central Kurdish** and **Northern Kurdish (Kurmanji)** were not included in some of the language-specific experiments because several evaluated models did not provide adequate support for these languages. Furthermore, Central Kurdish and Northern Kurdish are written using different scripts in the datasets considered. Central Kurdish is predominantly represented using the **Arabic-based Sorani script**, whereas Northern Kurdish (Kurmanji) is predominantly represented using the **Latin-based script**. This difference in writing systems can introduce additional challenges for multilingual speech models, particularly in transcription and text-based evaluation.

4.1 Arabic

- **Qwen/Qwen3-ASR-0.6B**: ASR-only model. WER was approximately 0.682 and CER was 0.271. The CER was considerably lower than the WER, indicating that a substantial portion of the errors occurred at the word level while character-level similarity was partially preserved.

4.2 Persian

- **facebook/seamless-m4t-v2-medium**: A multimodal speech foundation model capable of transcription and speech translation. WER and CER were 0.491 and 0.279 respectively. Transcription was reasonably strong, but many sentences were not completely translated and semantic meaning was limited.
- **openai/whisper-large-v3**: WER and CER were 0.434 and 0.213. Although one or two samples exhibited extreme errors, including WER and CER values as high as approximately 1.833 and 1.454, the overall transcription was substantially better. Translation quality was generally better than Seamless-M4T, although some semantic mismatches remained.
- **google/gemma-4-E2B-it**: The model supports both **ASR** and **speech translation**. As a prompt-based model, the same prompt was provided for all audio samples.

Task: Transcribe the source speech accurately. Do not skip any sentences. Translate the transcription into English. Do not skip any sentences.

IMPORTANT FORMAT (must follow exactly):

Source language: <transcription>

English: <translation>

The model achieved a WER and CER of approximately 0.446 and 0.243, respectively, comparable to **Facebook SeamlessM4T**. However, the translation quality was substantially weaker, with many sentences remaining untranslated or being skipped entirely. This discrepancy may result from the ASR and translation stages being evaluated separately and will be further investigated in the **Gemma 4 model-wise benchmark**.

- **Qwen/Qwen3-ASR-0.6B:** ASR-only model with WER 0.656 and CER 0.319. This was the weakest transcription result among the evaluated Persian models.

4.3 Georgian

- **facebook/seamless-m4t-v2-large:** WER 0.333 and CER 0.119. Transcription performance was strong and translations were close to the reference. The unusually strong result may raise a question about a possible training-data overlap.
- **google/gemma-4-E2B-it:** WER 0.546 and CER 0.203. Performance declined compared with Seamless-M4T, and some translation outputs were ignored.

4.4 Russian

- **Qwen/Qwen3-ASR-0.6B:** WER 0.330 and CER 0.087. The unusually low CER may indicate strong familiarity with the evaluation domain and warrants investigation of potential training-data overlap.
- **facebook/seamless-m4t-v2-large:** WER 0.364 and CER 0.102. Translation quality was strong. However, the audio files contained relatively few words per segment, motivating future evaluation using longer-form speech.

4.5 Somali

- **facebook/mms-1b-all:** WER 0.447 and CER 0.119, representing a strong ASR baseline.
- **google/gemma-4-E4B:** WER 0.547 and CER 0.200. Two out of twenty audio samples were transcribed using Arabic script instead of Somali script. Translation quality was also poor relative to the Google Translate reference.
- **ElevenLabs Scribe v2:** WER 0.375 and CER 0.114, giving the strongest transcription result among the models evaluated in this experiment. The unusually low error rate may warrant further investigation for potential training-data overlap.
- **facebook/seamless-m4t-v2-large:** WER 0.325 and CER 0.102. This was the lowest error rate among the evaluated models. Several generated translations were identical or nearly identical to the reference translations, motivating further investigation into possible data overlap.
- **google/gemma-4-E2B-it:** WER 0.512 and CER 0.165. Character-level errors were relatively limited compared with the word-level errors, but translation quality remained weaker than the other evaluated systems.

4.6 Tigrinya

- **facebook/seamless-m4t-v2-medium 1.2b**: Evaluated primarily for speech translation because ASR transcription output was not available for this language. The generated translations were partially meaningful, although several outputs were incorrect.
- **google/gemma-4-E2B-it**: WER 0.966 and CER 0.627. Transcription performance was poor, with several sentences partially translated or skipped entirely.

4.7 Turkish

- **Qwen/Qwen3-ASR-0.6B**: WER 0.185 and CER 0.051. The unusually low error rates suggest that possible training-data overlap should be investigated.
- **facebook/seamless-m4t-v2-large**: WER 0.277 and CER 0.097. Translation quality was strong, with several outputs closely matching the reference translations.

5 Model-Wise Systematic Benchmarks

5.1 TranslateGemma Benchmark

The TranslateGemma benchmark evaluated the 12B and 27B models across 19 languages: Tamil, Italian, Albanian, French, Irish, Romanian, Romansh, Russian, Malayalam, Spanish, Turkish, Northern Kurdish, Somali, Georgian, Tigrinya, Persian, Arabic, Luxembourgish, and Afghan Dari. The evaluation was performed using COMET and BLEU metrics, including unigram, bigram, trigram, and four-gram BLEU scores. Afghan Dari was evaluated using a text-to-text translation setup, as no corresponding audio benchmark was available.

Language	Res.	COMET	BLEU	1g	2g	3g	4g
Tamil	Low	0.843	0.217	0.488	0.267	0.166	0.102
Italian	Med-High	0.873	0.389	0.649	0.432	0.324	0.252
Albanian	Med	0.919	0.434	0.706	0.515	0.376	0.258
French	High	0.891	0.397	0.677	0.462	0.333	0.239
Irish	Low-Med	0.825	0.383	0.606	0.413	0.320	0.268
Romanian	Med	0.934	0.535	0.760	0.593	0.472	0.384
Romansh	Low	0.647	0.135	0.394	0.173	0.092	0.053
Russian	High	0.805	0.250	0.562	0.331	0.193	0.109
Malayalam	Med	0.872	0.274	0.553	0.339	0.218	0.138
Spanish	High	0.894	0.436	0.702	0.508	0.377	0.270
Turkish	Med	0.891	0.315	0.610	0.378	0.258	0.165
Northern Kurdish	Low	0.818	0.331	0.578	0.380	0.265	0.207
Somali	Low	0.819	0.193	0.491	0.246	0.135	0.085
Georgian	Low	0.868	0.270	0.557	0.335	0.212	0.134
Tigrinya	Low	0.757	0.114	0.422	0.143	0.071	0.039
Persian	Med	0.777	0.221	0.542	0.283	0.163	0.095
Arabic	Med-High	0.800	0.270	0.563	0.330	0.215	0.141

Language	Res.	COMET	BLEU	1g	2g	3g	4g
Luxembourgish	Low	0.647	0.150	0.461	0.195	0.100	0.056
Afghan Dari	Low	0.879	0.381	0.652	0.448	0.319	0.226

Table 2: TranslateGemma 12B Translation Performance

Language	Res.	COMET	BLEU	1g	2g	3g	4g
Tamil	Low	0.849	0.194	0.509	0.275	0.148	0.068
Italian	Med-High	0.865	0.223	0.422	0.256	0.177	0.129
Albanian	Med	0.928	0.535	0.764	0.615	0.477	0.364
French	High	0.891	0.397	0.677	0.462	0.333	0.239
Irish	Low-Med	0.817	0.405	0.610	0.454	0.355	0.275
Romanian	Med	0.942	0.585	0.790	0.635	0.530	0.440
Romansh	Low	0.638	0.129	0.393	0.169	0.088	0.047
Russian	High	0.802	0.295	0.593	0.374	0.237	0.145
Malayalam	Med	0.852	0.267	0.504	0.327	0.211	0.146
Spanish	High	0.868	0.378	0.672	0.456	0.315	0.211
Turkish	Med	0.895	0.363	0.627	0.420	0.308	0.215
Northern Kurdish	Low	0.822	0.357	0.627	0.426	0.297	0.204
Somali	Low	0.822	0.357	0.627	0.426	0.297	0.204
Georgian	Low	0.871	0.291	0.594	0.357	0.231	0.147
Tigrinya	Low	0.774	0.162	0.470	0.207	0.113	0.062
Persian	Med	0.779	0.239	0.552	0.296	0.177	0.113
Arabic	Med-High	0.795	0.301	0.569	0.356	0.240	0.169
Luxembourgish	Low	0.660	0.174	0.495	0.230	0.119	0.068
Afghan Dari	Low	0.881	0.372	0.644	0.431	0.309	0.223

Table 3: TranslateGemma 27B Translation Performance

5.2 TranslateGemma Findings

Several broad trends were observed:

1. Low-resource languages such as Tamil, Romansh, Tigrinya, Luxembourgish, and Somali generally exhibited lower BLEU scores.
2. COMET was often substantially higher than BLEU for low-resource languages. This suggests that semantic similarity can remain relatively strong even when the exact word sequence differs from the reference.
3. Romanian and Albanian produced particularly strong results across both COMET and BLEU. Given the relatively high overlap, these results warrant additional investigation into potential training-data overlap.

4. The 27B model did not consistently outperform the 12B model. Improvements were language-dependent.
5. The difference between COMET and BLEU highlights the importance of using both semantic and surface-form evaluation metrics for multilingual translation.

6 Whisper Architectural Benchmark

The Whisper benchmark compared **whisper-large-v3** and **whisper-large-v3-turbo** across 20 languages.

Language	Code	WER	CER	MER	WIL
Albanian	sq	0.622	0.183	0.578	0.801
Central Kurdish	ckb	1.431	0.575	0.956	0.996
English	en	0.163	0.054	0.162	0.265
French	fr	0.266	0.085	0.261	0.202
Irish	ga	1.036	0.608	0.930	0.993
Italian	it	1.974	1.232	0.957	0.996
Northern Kurdish	kmr	1.019	0.851	0.995	0.999
Romanian	ro	0.105	0.021	0.105	0.195
Romansh	rm	0.954	0.459	0.878	0.982
Russian	ru	0.236	0.061	0.235	0.392
Spanish	es	0.202	0.081	0.200	0.328
Tamil	ta	0.557	0.138	0.541	0.765
Turkish	tr	0.622	0.183	0.578	0.801
Malayalam	ml	3.860	2.322	1.000	1.000
Somali	so	0.877	0.289	0.804	0.956
Georgian	ka	0.727	0.217	0.709	0.904
Tigrinya	ti	2.906	2.636	1.000	1.000
Persian	fa	0.430	0.212	0.387	0.547
Arabic	ar	0.052	0.021	0.052	0.087
Luxembourgish	lb	0.755	0.443	0.621	0.819

Table 4: Whisper Large-v3 Performance

Language	Code	WER	CER	MER	WIL
Albanian	sq	0.719	0.241	0.677	0.884
Central Kurdish	ckb	1.322	0.821	0.968	0.998
English	en	0.170	0.058	0.168	0.275
French	fr	0.274	0.087	0.269	0.425

Language	Code	WER	CER	MER	WIL
Irish	ga	1.040	0.681	0.980	0.999
Italian	it	0.694	0.251	0.617	0.834
Northern Kurdish	kmr	3.603	2.551	1.000	1.000
Romanian	ro	0.109	0.022	0.109	0.203
Romansh	rm	0.981	0.374	0.893	0.986
Russian	ru	0.236	0.063	0.235	0.389
Spanish	es	0.231	0.083	0.229	0.380
Tamil	ta	0.666	0.180	0.647	0.858
Turkish	tr	0.085	0.020	0.084	0.152
Malayalam	ml	2.473	1.400	1.000	1.000
Somali	so	1.360	0.560	0.916	0.988
Georgian	ka	1.000	1.122	1.000	1.000
Tigrinya	ti	2.025	2.217	1.000	1.000
Persian	fa	0.382	0.159	0.364	0.541
Arabic	ar	0.121	0.043	0.119	0.210
Luxembourgish	lb	0.417	0.241	0.397	0.594

Table 5: Whisper Large-v3-Turbo Performance

6.1 Whisper Observations

The Whisper benchmark demonstrates substantial language-dependent variation. The strongest results were observed for languages such as Arabic, Romanian, English, French, Russian, and Spanish. In contrast, Malayalam, Tigrinya, Central Kurdish, Northern Kurdish, Irish, and Romansh produced substantially higher error rates. The Turbo model produced substantial improvements for some languages, most notably Italian and Turkish, while degrading performance for other languages such as Northern Kurdish and Somali. This indicates that architectural compression or decoding changes do not necessarily produce uniform improvements across languages.

7 Gemma 4 12B Prompt Evaluation

The Gemma 4 benchmark used a standardized prompt across all audio files and languages.

7.1 Prompt Configuration

```

You are a speech-to-text and translation system.
Task:
1. Transcribe the [SOURCE LANG] speech accurately.
2. Translate the transcription into English.
Output format:
SOURCE LANG:
English:
Rules:
Do NOT add extra text.
Do NOT summarize.
Keep sentence order same as speech.
```

Language	WER	CER	MER	WIL	Translation Quality / Observations
Albanian	1.122	0.641	0.808	0.947	Mixed performance; some translations were good, while others were affected by transcription errors or were omitted.
Central Kurdish	0.907	0.327	0.851	0.972	Some individual words were correct, but sentence-level semantic alignment was poor.
English	0.175	0.059	0.173	0.282	Transcription was near perfect.
French	0.294	0.137	0.289	0.428	Almost all translations were correct.
Irish	0.968	0.647	0.861	0.975	Most translations were incorrect, indicating substantial degradation for this low-to-medium-resource language.
Italian	1.414	0.644	0.989	0.999	Most translations were incorrect; some Italian words remained untranslated in the English output.
Northern Kurdish	0.910	0.349	0.872	0.979	Most transcriptions were incorrect and translations did not match the reference semantically or lexically.
Romanian	0.570	0.236	0.548	0.776	Approximately half of the sentences were translated correctly; the remainder were largely out of context.
Romansh	0.818	0.295	0.779	0.947	Most translations were incorrect and some source-language words remained untranslated.
Russian	0.386	0.154	0.358	0.540	Translations were very strong and may warrant investigation for possible training-data overlap.
Spanish	0.300	0.156	0.285	0.436	Most translations were correct, with localized semantic errors.
Tamil	1.078	0.503	0.946	0.996	Most transcriptions were incorrect, leading to incorrect final translations.
Turkish	0.886	0.498	0.798	0.950	One sentence was transcribed in Hindi/Devanagari despite Turkish being specified as the source language. Translation quality was poor.
Malayalam	2.106	0.968	0.948	0.994	Only a few sentences were translated correctly; most were out of context.
Somali	1.155	0.886	0.894	0.983	Most translations were incorrect and many sentences were not translated.
Georgian	2.278	1.567	0.964	0.996	Outputs showed severe failure and did not resemble the reference sentences.
Tigrinya	1.027	0.754	0.974	0.999	Severe transcription errors and character looping were observed. Translation output was largely absent.

Language	WER	CER	MER	WIL	Translation Quality / Observations
Persian	0.962	0.830	0.959	0.994	Many sentences were not transcribed, potentially related to long input sequences. Translation was consequently incomplete.
Arabic	0.373	0.126	0.361	0.570	Most translations preserved the meaning reasonably well despite differences in word order.
Luxembourgish	0.984	0.494	0.858	0.976	Approximately half the sentences were translated correctly; others were incomplete or out of context.

Table 6: Gemma 4 12B Transcription and Translation Evaluation

8 Omnilingual ASR Benchmark

The OmniASR models were evaluated using six different configurations:

1. omniASR_3B_CTC
2. omniASR_7B_CTC
3. omniASR_3B_LLM
4. omniASR_7B_LLM
5. omniASR_3B_LLM_Unlimited_V2
6. omniASR_7B_LLM_Unlimited_V2

For OmniASR models requiring audio segments shorter than approximately 40 seconds, longer audio files were segmented before inference.

Language	WER	CER	MER	WIL	Observations
Tamil	0.431	0.082	0.424	0.631	One sentence was transcribed in Hindi/Devanagari.
Albanian	0.566	0.208	0.563	0.796	Two sentences were transcribed in Arabic script.
Arabic	0.234	0.074	0.233	0.372	Consistent transcription performance.
Turkish	0.362	0.133	0.359	0.579	—
Somali	0.397	0.108	0.377	0.585	—
Russian	0.359	0.089	0.356	0.557	—
Luxembourgish	0.949	0.414	0.876	0.981	High error rate.
Georgian	0.130	0.033	0.129	0.217	Strong result.
Central Kurdish	0.387	0.075	0.379	0.589	—
English	0.231	0.071	0.227	0.373	—
French	0.404	0.225	0.384	0.537	Audio longer than 40 seconds was segmented.
Persian	0.887	0.698	0.875	0.941	High transcription error.
Irish	0.663	0.240	0.640	0.858	—

Language	WER	CER	MER	WIL	Observations
Italian	0.466	0.078	0.463	0.711	–
Malayalam	0.393	0.061	0.373	0.575	–
Northern Kurdish	0.376	0.085	0.363	0.570	–
Romanian	0.328	0.061	0.328	0.547	–
Romansh	0.496	0.103	0.485	0.710	–
Spanish	0.299	0.099	0.296	0.484	–
Tigrinya	0.926	0.617	0.844	0.963	High error rate.

Table 7: OmniASR 3B-CTC Benchmark Performance

Language	WER	CER	MER	WIL	Observations
Tamil	0.429	0.080	0.421	0.628	One sentence was transcribed in Hindi/Devanagari.
Albanian	0.607	0.229	0.589	0.815	One sentence was transcribed in Arabic script.
Arabic	0.275	0.102	0.272	0.421	–
Turkish	0.298	0.083	0.297	0.503	–
Somali	0.391	0.106	0.373	0.584	–
Russian	0.332	0.083	0.330	0.525	–
Luxembourgish	0.923	0.384	0.852	0.975	High error rate.
Georgian	0.090	0.022	0.089	0.145	Strong result.
Central Kurdish	0.295	0.052	0.290	0.465	–
English	0.229	0.070	0.225	0.375	–
French	0.424	0.221	0.401	0.570	Audio longer than 40 seconds was segmented.
Persian	0.881	0.692	0.842	0.938	High error rate.
Irish	0.616	0.241	0.601	0.827	–
Italian	0.492	0.080	0.489	0.735	–
Malayalam	0.246	0.033	0.241	0.393	Strong improvement over 3B CTC.
Northern Kurdish	0.341	0.075	0.338	0.546	–
Romanian	0.305	0.055	0.305	0.518	–
Romansh	0.412	0.080	0.405	0.624	–
Spanish	0.299	0.098	0.297	0.482	–
Tigrinya	0.940	0.615	0.868	0.976	High error rate.

Table 8: OmniASR 7B-CTC Benchmark Performance

Language	WER	CER	MER	WIL	Observations
Tamil	0.373	0.079	0.367	0.563	–
Albanian	0.653	0.179	0.640	0.857	–
Arabic	0.219	0.093	0.217	0.352	–
Turkish	0.298	0.065	0.293	0.492	–
Somali	0.380	0.105	0.363	0.573	–
Russian	0.310	0.081	0.308	0.494	–
Luxembourgish	0.966	0.402	0.867	0.979	High error rate.
Georgian	0.080	0.027	0.080	0.128	Very strong result.
Central Kurdish	0.234	0.047	0.233	0.393	–
English	0.217	0.072	0.214	0.351	–
French	0.399	0.328	0.392	0.572	Audio longer than 40 seconds was segmented.
Persian	0.855	0.597	0.769	0.923	High error rate.
Irish	0.544	0.200	0.525	0.756	–
Italian	0.466	0.081	0.461	0.707	–
Malayalam	0.193	0.032	0.190	0.332	Strong result.
Northern Kurdish	0.292	0.070	0.290	0.486	–
Romanian	0.319	0.056	0.319	0.537	–
Romansh	0.378	0.084	0.375	0.596	–
Spanish	0.288	0.090	0.287	0.475	–
Tigrinya	1.225	0.744	0.885	0.981	Poor performance.

Table 9: OmniASR 3B-LLM Benchmark Performance

Language	WER	CER	MER	WIL	Observations
Tamil	0.331	0.071	0.325	0.503	–
Albanian	0.688	0.176	0.678	0.883	–
Arabic	0.183	0.067	0.179	0.306	Strong result.
Turkish	0.266	0.059	0.263	0.449	–
Somali	0.380	0.104	0.362	0.569	–
Russian	0.310	0.079	0.308	0.504	–
Luxembourgish	0.940	0.384	0.864	0.979	High error rate.
Georgian	0.033	0.006	0.033	0.064	Exceptionally strong result.
Central Kurdish	0.300	0.057	0.292	0.474	–
English	0.216	0.069	0.214	0.355	–
French	0.373	0.203	0.351	0.507	Audio longer than 40 seconds was segmented.
Persian	0.647	0.366	0.550	0.734	Improved relative to 3B LLM.
Irish	0.549	0.224	0.535	0.771	–

Language	WER	CER	MER	WIL	Observations
Italian	0.564	0.110	0.561	0.807	–
Malayalam	0.253	0.044	0.248	0.412	–
Northern Kurdish	0.287	0.069	0.284	0.477	–
Romanian	0.296	0.054	0.296	0.505	–
Romansh	0.382	0.076	0.375	0.595	–
Spanish	0.288	0.088	0.284	0.470	–
Tigrinya	1.031	0.595	0.788	0.940	Poor performance.

Table 10: OmniASR 7B-LLM Benchmark Performance

Language	WER	CER	MER	WIL	Observations
Tamil	0.438	0.161	0.430	0.612	–
Albanian	0.576	0.149	0.562	0.796	–
Arabic	0.248	0.090	0.241	0.393	–
Turkish	0.274	0.057	0.270	0.457	–
Somali	0.387	0.107	0.368	0.578	–
Russian	0.298	0.072	0.296	0.483	–
Luxembourgish	0.943	0.386	0.864	0.979	High error rate.
Georgian	0.069	0.052	0.068	0.125	Very strong result.
Central Kurdish	0.333	0.065	0.324	0.518	–
English	0.219	0.074	0.217	0.354	–
French	0.250	0.090	0.246	0.392	–
Persian	0.534	0.329	0.460	0.623	Improved over standard 3B LLM.
Irish	0.668	0.298	0.632	0.847	One sentence was transcribed in Japanese.
Italian	0.518	0.090	0.505	0.748	–
Malayalam	0.340	0.062	0.326	0.519	–
Northern Kurdish	0.306	0.070	0.305	0.505	–
Romanian	0.319	0.058	0.319	0.537	–
Romansh	0.564	0.181	0.535	0.764	–
Spanish	0.286	0.086	0.283	0.466	–
Tigrinya	0.727	0.420	0.568	0.759	Improved over standard 3B LLM.

Table 11: OmniASR 3B-LLM-Unlimited v2 Benchmark Performance

Language	WER	CER	MER	WIL	Observations
Tamil	0.411	0.140	0.403	0.586	Audio batches 25–26 failed; metric computed on successful samples.
Albanian	0.545	0.137	0.537	0.776	–
Arabic	0.201	0.073	0.198	0.335	–
Turkish	0.263	0.054	0.260	0.447	–
Somali	0.380	0.103	0.365	0.576	–
Russian	0.298	0.070	0.296	0.483	–
Luxembourgish	0.943	0.387	0.864	0.979	High error rate.
Georgian	0.009	0.001	0.009	0.017	Extremely strong result; requires careful validation for possible data overlap.
Central Kurdish	0.377	0.065	0.365	0.567	–
English	0.211	0.070	0.209	0.343	–
French	0.234	0.081	0.228	0.367	–
Persian	0.467	0.279	0.402	0.540	Strongest Persian result among the reported OmniASR variants.
Irish	0.523	0.191	0.507	0.743	–
Italian	0.533	0.108	0.525	0.770	–
Malayalam	0.220	0.033	0.217	0.366	Strong result.
Northern Kurdish	0.287	0.067	0.285	0.479	–
Romanian	0.292	0.052	0.292	0.499	–
Romansh	0.397	0.089	0.394	0.620	–
Spanish	0.296	0.092	0.293	0.480	–
Tigrinya	0.715	0.415	0.553	0.738	Best reported Tigrinya result among the OmniASR variants.

Table 12: OmniASR 7B-LLM-Unlimited v2 Benchmark Performance

9 Key Findings and Limitations

Across the language-specific and model-wise experiments, several recurring patterns were observed.

9.1 Dataset and Training-Data Overlap

Several evaluations produced unexpectedly strong results, particularly on low-resource datasets. Examples include very low WER/CER values for Georgian, Turkish, Russian, and Somali in some model configurations. These results should not be interpreted as definitive evidence of dataset contamination. However, the combination of unusually low error rates, high n -gram overlap, and in some cases nearly identical translations suggests that training-data overlap should be investigated. A stronger validation procedure would include:

1. Evaluating on a newly collected legal ASR and text-to-text dataset.
2. Testing utterances unavailable in publicly released benchmarks.
3. Comparing performance against multiple independent datasets.

9.2 Long-Form Speech Limitations

Several models exhibited degradation when processing longer audio segments. This was especially visible in Gemma 4 and some speech translation configurations. Potential failure modes included:

- Sentence ignorance;
- Incomplete transcription;
- Translation termination;
- Repeated characters or tokens;
- Partial translations.

For OmniASR models requiring shorter inputs, audio segments exceeding approximately 40 seconds were split before inference.

9.3 COMET–BLEU Metric Disagreement

One of the most important findings from the TranslateGemma benchmark was the difference between semantic and surface-form metrics. For example, Tamil achieved a COMET score of 0.843 with an overall BLEU score of only 0.217 using TranslateGemma 12B. Similarly, Somali achieved a COMET score of 0.819 while obtaining a BLEU score of 0.193. This indicates that a translation can preserve much of the semantic meaning while differing substantially from the reference in exact word ordering and lexical realization. Therefore, evaluating low-resource translation systems using BLEU alone may underestimate their semantic quality.

10 Comparative Model Observations

10.1 Whisper

Whisper showed strong performance on several high-resource languages but displayed substantial degradation on several low-resource languages, including Malayalam, Tigrinya, Northern Kurdish, and Romansh. The Turbo model improved substantially over Large-v3 for certain languages, including Italian and Turkish, but did not provide consistent improvements across the complete language set.

10.2 Gemma 4

Gemma 4 demonstrated strong performance for several high-resource languages such as English and French. However, performance degraded considerably for low-resource languages. The direct prompt-based approach also revealed a distinct failure mode: although the model was explicitly instructed to perform both transcription and translation, it sometimes completed transcription while omitting the translation stage.

10.3 OmniASR

The OmniASR experiments showed that scaling from 3B to 7B parameters did not universally improve performance. The LLM-based variants generally showed stronger performance than the CTC variants for several languages, although the gains varied substantially. The strongest individual results were observed for Georgian in the 7B LLM and 7B LLM Unlimited configurations. Such unusually strong results should be validated on independent test sets before drawing conclusions about generalization.

11 Future Work

The following directions could strengthen the benchmark and help identify an effective multilingual speech-processing pipeline under a total model-size constraint of approximately 25 GB:

1. **Evaluate models on legal-context texts.** Testing the models on legal-domain texts to determine whether their transcription and translation capabilities remain reliable for domain-specific terminology, long sentences, and legally sensitive content.
2. **Evaluate speech-to-text models as alternatives to text-to-text translation models.** Instead of relying exclusively on an ASR followed by a text-to-text translation pipeline, investigate whether speech-to-text models can directly produce accurate translated text from speech.
3. **Evaluate speech-to-speech models.** Investigate whether end-to-end speech-to-speech models can provide accurate and natural translations while reducing the need for separate ASR and translation components.
4. **Identify the optimal model combination under a 25 GB constraint.** Systematically compare different model combinations, including Whisper variants for high- and low-resource languages, OmniASR combined with TranslateGemma, and larger unified models such as Gemma 4 or other multilingual speech models. The objective is to determine whether a modular pipeline provides better language coverage and accuracy than a single larger model within the same memory budget.

5. **Compare modular and unified architectures.** Compare alternatives such as Whisper + translation, OmniASR + TranslateGemma, a single unified speech model, and speech-to-speech architectures to determine which approach provides the best trade-off between model size, language coverage, transcription accuracy, translation quality, and computational cost.

12 Conclusion

This internship involved a comprehensive evaluation of modern multilingual ASR and speech translation systems across 21 languages, with particular emphasis on low-to medium-resource languages. The results demonstrated substantial variation in performance across languages and model architectures, highlighting the influence of script representation, training-data coverage, and input characteristics on transcription and translation quality. The experiments also identified important failure modes, including cross-script hallucinations, sentence omission, incomplete translation, character looping, and severe degradation in several low-resource settings. Overall, this benchmark provides a foundation for systematic evaluation of multilingual speech models and offers valuable insights into their strengths and limitations. This work will be continued through further experiments and detailed analysis to develop the study into a valuable research outcome.